

Pair Entropy Is $2h(\alpha_A) + 2h(\alpha_B)$. δS Tracks Mass. $\nabla \cdot \mathbf{g}$ Is Not Poisson.

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Abstract

The orthonormal two-source state has four moving occupations $\alpha_A, 1 - \alpha_A, \alpha_B, 1 - \alpha_B$. Hence

$$S_{\text{global}} = 2h(\alpha_A) + 2h(\alpha_B).$$

The 4×4 Gram and C confirm this to 10^{-80} . Exact open-hop basis, no hop LAPACK. Peschel on each $R = 3$ ball is `mpmath eigsy` at 80 decimals.

64 balls: $\rho(f_S, f_E) = 0.957$, RMS 0.043. The one ball that contains both source *sites* still leaks ($f_S = 0.27$ below 1). Auditor $N = 11$: tracking confirmed; both-inside refuted.

Lattice $\nabla \cdot \mathbf{g}$ of the Paper 70 field is not a source law. $N = 12$ source/mid ratio 3.83 passes the pre-register; $N = 11$ finds the midpoint larger. Signs mix. $\mathbf{g}$ is a ball probe, not Poisson. Not Einstein.

Admission. *I asked $f = 1$ on a both-site ball and $|\nabla \cdot \mathbf{g}|$ to mark the packets. Both failed as Newton laws. I did not move them.*